



SMART METER

JINGZHANG UTILITY

Building intelligence as the fourth utility

Along 9 km of the Jingzhang Railway Heritage Park, Haidian, Beijing

Water 1908

Power 1888

Gas/Heat 1950s

Intelligence 2026

Beijing's piped water began in 1908 · electric light in 1888 (Xiyuan) · district heating in the 1950s · the smart utility begins with the Centennial Jingzhang AI Innovation Belt

Centennial Jingzhang AI Innovation Belt — urban design concept · submission jingzhang-rail-lab · author miaotony (GitHub)

Coordinated study 43.6 km² · overall design 11.4 km² · key areas 368.4 ha · all recomputed on provisional boundaries in EPSG:4548

Two grid connections in one century: from railway to smart utility network

1909 — first connection

The Jingzhang (Beijing–Zhangjiakou) Railway plugged Beijing into a railway network designed and operated by Chinese engineers. Over the following century, every step of urban modernization was in essence "utilitization": piped water into homes, electric light onto desks, heating into buildings.

2026 — second connection

Beneath 9 km of the Jingzhang Railway Heritage Park, a smart utility tunnel — iDUCT — delivers compute, models and data into every building, every small shop and every community, just like water and power.

Three tests of a fourth utility

Meterable: the Watt-Token joint bill puts kWh and inference volume on one statement.
Settleable: the Three Ledgers are publicly reconciled every quarter.
Universal service: seniors, small shops and schools get an affordable tier.

Why these 9 km

AI core industry worth CNY 350 bn, 30% of the national total. 2,000+ companies, 26 unicorns, 130 registered large models. The corridor concentrates 70% of Haidian enterprise revenue and 80% of its AI talent. 37 universities, 52 national key laboratories, 12,300 AI scholars.

Official progress anchors

Heritage Park Phase II fully opens end of 2026 (530,900 m²). Xuebei Park 238,300 m², opening July 2026. Origin Tower renamed March 2026, 439 companies in residence. Usable renewal floor area 1,000,000 m² (official public figure).

What sets this apart from 309 entries

This proposal stops treating the AI belt as a "testbed" or "demonstration zone" and builds it as a utility: with a balance sheet (Space Ledger), metering and settlement (Watt-Token Ledger), rights confirmation and circulation (Data Ledger), and a universal-service obligation (three-tier pricing).

Concept statement

All mechanisms are conceptual proposals marked "pending organizer confirmation"; all geometry is based on provisional boundaries and will be recomputed once official red lines are released.

A 9 km smart utility tunnel (iDUCT) links Source Plant - Transformer - Metering District, like generation-transmission-distribution-consumption of a city power grid

Zhongzhi Park = Source Plant
192.1 ha

SOURCE PLANT
models & standards,
generation side

AI Origin Community = Transformer
104.3 ha

TRANSFORMER
university origination,
step-up to grid

METERING DISTRICT
billing side · data trading

Dazhongsì = Metering District
72.0 ha

Qinghua East Rd W.
Service Hall

Wudaokou Service Hall
talent first stop

Dazhongsì Service Hall
four-quadrant connection

▼ Xizhimen · city gateway
1 km

- Energy trunk (green power + waste heat)
- Data / compute trunk
- Heritage-park green corridor (iDUCT surface)
- Plant-station node
- Service Hall (integrated with metro)
- Key area (provisional boundary)

JINGZHANG UTILITY · Fig.1 One Duct, Three Plant-Stations — over Provisional boundary, not official red line; statutory plans prevail

One Duct, Three Plant-Stations

The 9 km iDUCT works like the trunk line of a city power grid: the northern Source Plant generates (models and standards), the central Transformer steps up (university origination onto the grid), the southern Metering District bills (smart economy and data trading), and three metro stations are upgraded into Service Halls.

Source Plant · Zhongzhi Park 192.1 ha

The "generation side" of models and standards. Anchor: Xuebei Park, 238,300 m², opening July 2026.

Transformer · AI Origin Community 104.3 ha

University research "stepped up onto the grid". Anchors:
Origin Tower, Tsinghua AI College Building F, the 1 km
conversion circle.

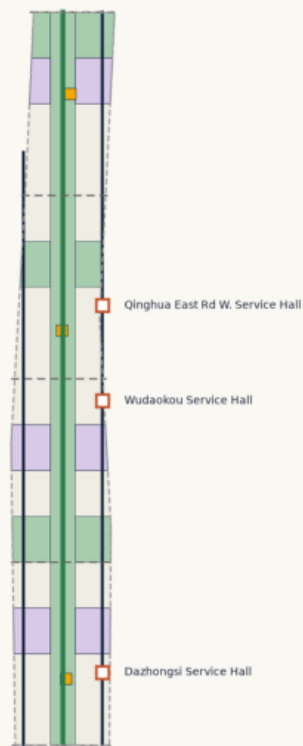
Metering District · Dazhongsi 72.0 ha

Billing side + data-element exchange. Echoes the announcement on "data element and digital asset circulation mechanisms".

Three Service Halls

Dazhongsi (four-quadrant connection), Wudaokou (talent first stop), Qinghua East Road West (campus stitching).

Three underground chambers: energy (distributed green power + storage), compute (edge inference nodes, hot-swappable every 300 m), pipeline (data trunk + reclaimed-water cooling). The surface is a park — and the perceptible interface of the utility.



Compute waste heat warms pools and greenhouses, melts snow on winter paths and feeds hot-spring mist gardens (order-of-magnitude estimates in assumptions).

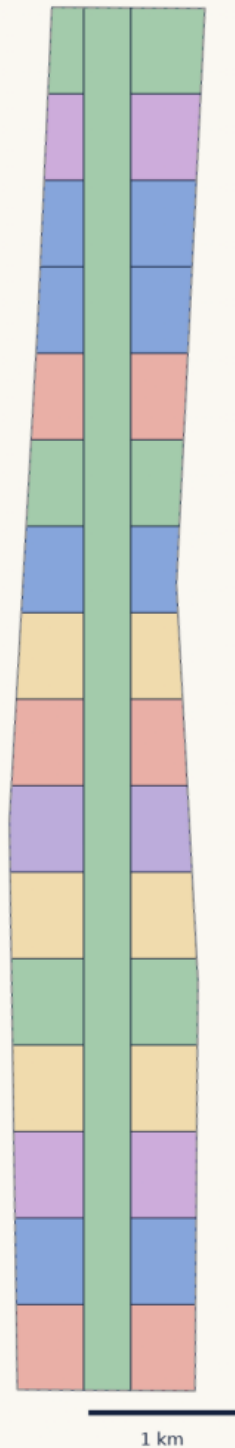
Citizens watch the compute chamber run through glass hatches — "seeing the city think".

A light strip along the 9 km trail displays green-power share and compute load in real time (conceptual demo).

"Explore the integration of distributed energy, edge compute and other new service facilities with the three traditional utilities" — this proposal turns that sentence into space.

Space Ledger · 32 Parcel Accounts

Each renewal parcel = one account: land-use type x demolish/retrofit/retain x rent tier;
official anchor = 1,000,000 m² of usable renewal floor area



Account land-use types (32 parcels, full coverage · 0.000% error)					
R&D land	Commercial services	Cultural facilities			
Residential	Public services	Green / open space			

JINGZHANG UTILITY · Fig.2 Space Ledger parcel accounts

Provisional boundary, not official red line; statutory plans prevail

32 parcel accounts

Within 11.4 km², each renewal parcel = one account: assets (existing buildings and land), input (renewal funding), output (rent and services), obligations (universal-service quota). The demolish/retrofit/retain triage is anchored to the official "1,000,000 m² of usable renewal floor area".

Rent gradient

Source Plant: low rent, long leases (40–60% of market, milestone-stepped); Metering District: market rent +15–20% premium paid into the universal-service fund.

FAR transfer pool

Development rights transferred from stock parcels concentrate in the four Dazhongsi quadrants, supporting over-station development at the Service Hall.

Renewal project list

proposal.md appends a 10-row concept list: project / location / account type / funding source / payback period.

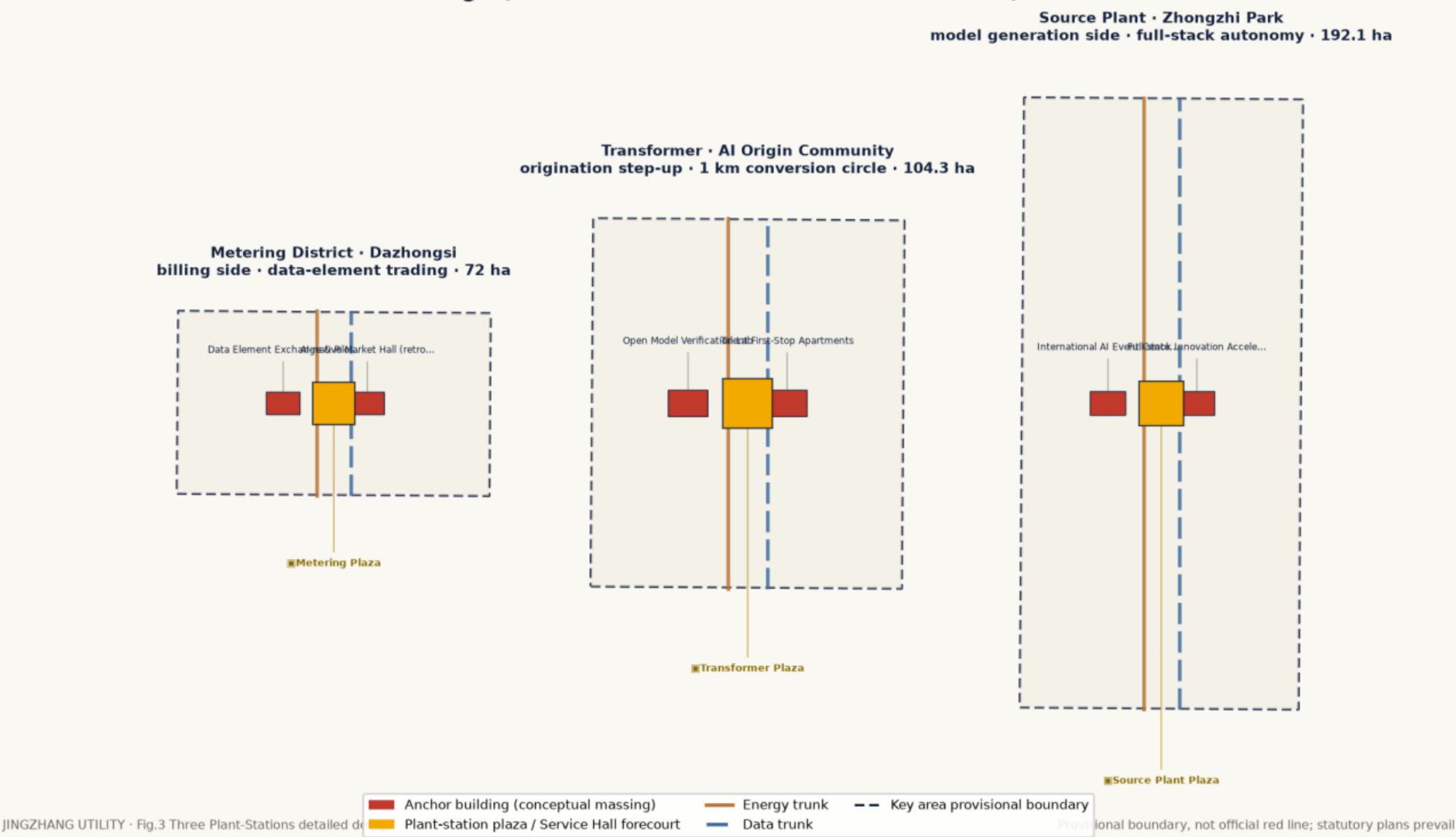
Regulatory-depth response

The brief asks for a "renewal implementation project list and policies" — the account system turns it into an auditable table.

Three Plant-Stations — detailed design

Spatializing generate-transmit-distribute-consume: each plant-station combines anchor buildings (red) + a plant-station plaza / Service Hall forecourt (yellow) + the twin energy and data trunks.

Three Plant-Stations · Detailed Design (Generate-Transmit-Distribute-Consume)



Metering District first (yrs 1-3)

Dazhongsi Service Hall + first iDUCT segment; use the metro-station retrofit window to prove metering and settlement.

Transformer networked (yrs 2-5)

Origin Community Service Hall + central segment, meshing with the Heritage Park Phase II opening at end of 2026.

Source Plant at full output (yrs 3-8)

Zhongzhi Park full-stack clusters + Three Ledgers running corridor-wide, aligned with the industrial ramp-up after Xuebei Park opens.

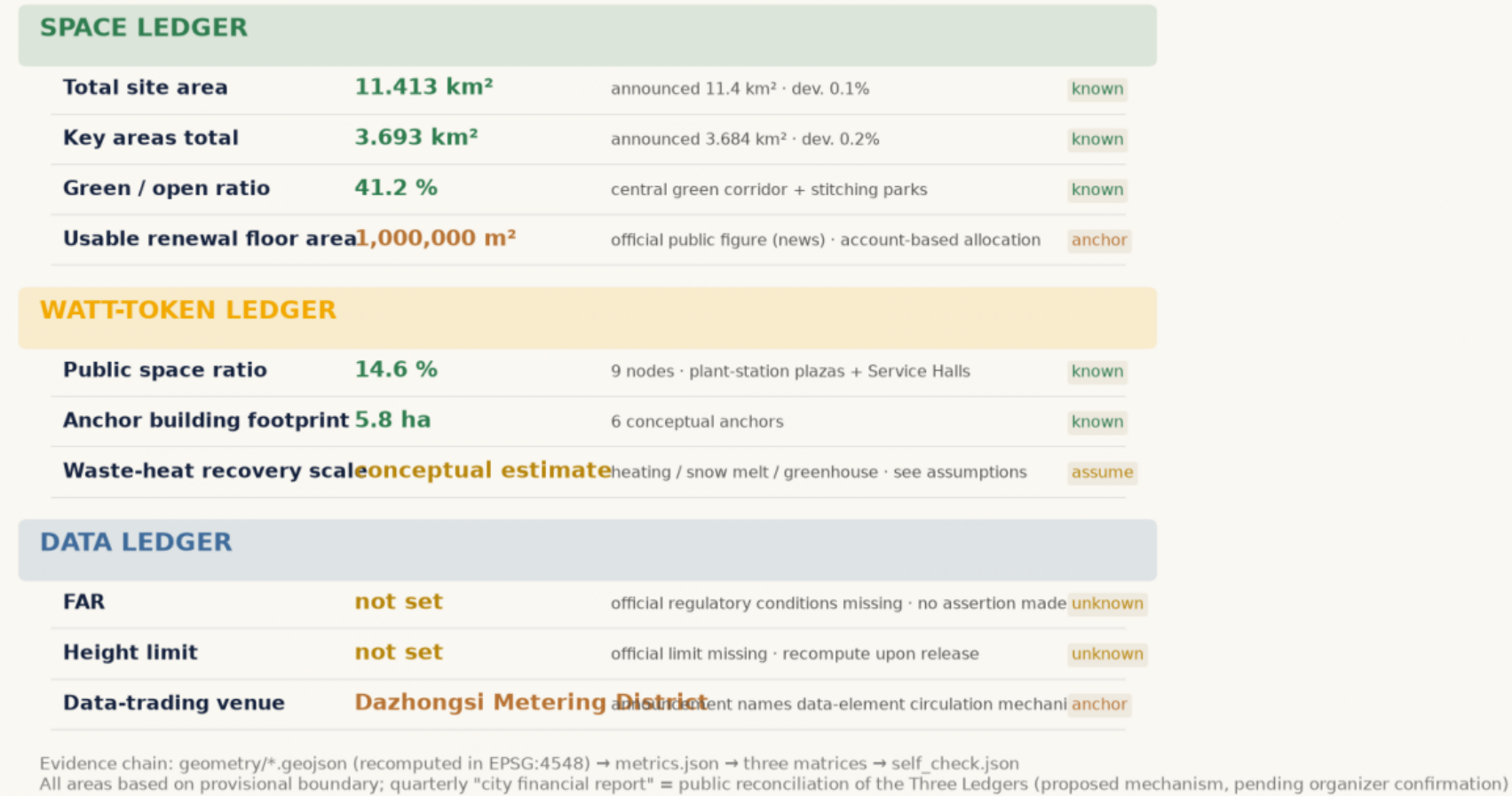
Blue-green baseline

Green/open space 41.2%, public space 14.6%; the Qinghe and Xiaoyuehe ecological interfaces are stitched along the whole line.

Three City Ledgers & the evidence chain

Space Ledger, Watt-Token Ledger and Data Ledger are publicly reconciled every quarter — a city financial report. All metrics independently recomputed in EPSG:4548, within 0.2% of announced figures.

Three City Ledgers · Metrics Evidence Chain



Space Ledger

Site 11.413 km² (announced 11.4, dev. 0.1%); key areas 3.693 km² (dev. 0.2%).

Watt-Token Ledger

A joint bill metering kWh and inference volume on one statement; bills redeem compute and model vouchers (linked to Beijing policy).

Data Ledger

Dazhongsi data exchange: rights confirmation / balance-sheet entry / circulation in one hall; tiered data gates; personal data trusts.

Evidence chain

geometry → metrics → three matrices → self_check; fully recomputable, open data.

JINGZHANG UTILITY · Fig.5 Three Ledgers metrics evidence chain

Provisional boundary, not official red line; statutory plans prevail

Ten door-to-door smart services · three-tier pricing · compliance statement

Door-to-door services (part 1)

- 1. Smart commuting: Service Hall shuttle buses in dynamic formation
- 2. Neighborhood care: community care pods (affordable tier)
- 3. Boundless learning paths: the park as classroom
- 4. Smart shop books: AI bookkeeping and stocking for small shops
- 5. Job-seeking origin: AI interview pods (already piloted officially)

(part 2)

- 6. Repair everything: AI repair workshops
- 7. Green-corridor stewards: landscaping robot fleet (already operating officially)
- 8. Legal beacon: AI legal aid
- 9. Silver companionship: companion terminals in communities
- 10. Emergency line guard: corridor-wide AI safety drills

Three-tier pricing

Affordable tier: free, covered by the universal-service fund (seniors, schools, micro-merchants).
Standard tier: pay-as-you-go (Watt-Token joint bill).
Industrial tier: market price; the premium cross-subsidizes the affordable tier.

Operating entity

Jingzhang Utility Co. (SPV): tripartite governance by government, enterprises and citizen representatives; the citizen seat holds a veto over universal-service terms. Universal-service fund: Metering District rent premium + industrial-tier share.

Compliance statement

This is a conceptual proposal, not a statutory planning product. All spatial extents are based on the provisional boundary PROV-SITE-001, not official red lines; FAR and height limits are marked unknown because official conditions are missing. All mechanisms are "proposed, pending organizer confirmation". Official public figures are cited for textual cross-reference only.

Copyright

COMMUNITY-DISPLAY-ONLY
For community display and review exchange only
Author: miaotony (GitHub)
Generated by a Hermes Agent multi-subagent pipeline